
Accountable Neural Aggregation in Computational Pathology: From Paired-Acquisition Representations to Whole-Slide and Institutional Learning

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Every component’s evidence maturity is stated explicitly. Architecture claims bind to source and tests; empirical claims bind to immutable result artifacts; pending and prohibited statuses are explicit.

Abstract

Computational-pathology neural systems aggregate information across several levels: patches are transformed into representations, patch instances are aggregated into whole-slide predictions, and institutional updates are aggregated into collaborative models. At each level, nuisance variation, shortcut signals, optimization failures, and institutional imbalance can be preserved or amplified. This research program develops neural architectures, paired-data designs, institutional-weighting protocols, stress tests, and provenance controls for auditing and improving those aggregation stages.

At the representation level, we introduce Paired-Acquisition Neural Factorization (PA-NF), a paired-data architecture with an explicit tissue-oriented biological branch and an explicit acquisition branch trained on exact same-region multi-scanner pairs, together with strong linear baselines (centroid/QR, PCA, paired-linear), broken-pair controls, and synthetic identifiability studies. At the whole-slide level, we introduce TransnnMIL, a multibranch neural aggregation architecture family with canonical global correlation modeling, an nnMIL-style aggregation path, branch-token fusion, hierarchical spatial pooling, a topology branch, adaptive pruning, and coordinate-aware graph caching. At the institutional level, we develop PathologyFL, a pathology-specific federated-learning research framework, and FAIR-WEIGHTS-H, an auditable hybrid institutional-weighting protocol that separates training, validation, and monitoring signals.

These components are connected by benchmark infrastructure (PCam patch evaluation), whole-slide benchmarks (PANDA), multi-center benchmarks (CAMELYON17), and a fail-closed scientific-audit and provenance system that binds every active empirical claim to an immutable result artifact.

The evidence maturity differs sharply across components and is stated explicitly. Corrected paired-scanner experiments support partial structured separation and paired supervision, but not neural feature-space superiority: no neural feature-space increment over simple scanner-removal baselines was supported under the corrected fixed five-category estimand, B64 increased retrieval and scanner recoverability without category gain, and synthetic accessibility effects did not transport to corrected real category

accessibility. TransnnMIL is an implemented architecture whose repaired matched-control performance evaluation remains pending. PathologyFL and FAIR-WEIGHTS-H support executable pathology-specific federated research and auditable institutional weighting without demonstrated clinical or fairness superiority.

The shared contribution is a multi-level framework for building and auditing accountable neural aggregation in computational pathology, in which architecture, corrected empirical evidence, negative results, proposed protocols, and pending controlled validation are kept rigorously distinct.

1 Introduction

Computational-pathology neural systems aggregate information across several distinct levels. At the first level, image patches are transformed into representations, either by a frozen foundation model or by a task-trained encoder. At the second level, patch instances are aggregated into whole-slide predictions through multiple-instance learning. At the third level, institutional model updates are aggregated into collaborative models through federated learning. These levels are usually studied as separate engineering problems, yet the artifacts that threaten them are shared: scanner and stain variation, acquisition workflow differences, site and center identity, label noise, optimization failures, and institutional imbalance can all be preserved, shortcut-exploited, or amplified as information passes from one aggregation stage to the next.

This research program treats those aggregation stages as a connected system and develops the neural architectures, paired-data designs, institutional-weighting protocols, stress tests, and provenance controls needed to build and audit them. The program spans:

- **Representation formation.** Patch and foundation-model representations and the scanner, stain, acquisition, site, region, and dataset artifacts they encode, studied through exact same-region paired scanner designs, Paired-Acquisition Neural Factorization (PA-NF), SCORPION and multi-scanner canine SCC, scanner and center subspaces, strong linear baselines, broken-pair controls, and synthetic identifiability studies.
- **Whole-slide aggregation.** TransnnMIL, a multibranch neural aggregation architecture family with canonical global

correlation modeling, an nnMIL-style aggregation path, branch-token fusion, hierarchical spatial pooling, a topology branch, adaptive pruning, and coordinate-aware graph caching, evaluated on PANDA feature bags.

- **Institutional aggregation.** PathologyFL, a pathology-specific federated-learning research framework, and FAIR-WEIGHTS-H, an auditable hybrid institutional-weighting protocol, exercised through dominant-site stress tests, corruption and label-noise stress, ordinal-shift detection, client heterogeneity and dropout, communication accounting, privacy probes, and CAMELYON17 center studies.
- **Benchmark and provenance infrastructure.** PCam patch-level evaluation, PANDA and CAMELYON17 whole-slide and center evaluation, and a fail-closed scientific-audit system that binds every active empirical claim to an immutable result artifact and every architectural claim to source and tests.

A central methodological position of the program is that *architecture, corrected empirical evidence, negative results, proposed protocols, and pending controlled validation* must be kept rigorously distinct. An architecture can be an authored contribution, implemented and tested, even when its superiority over matched baselines has not been established or has been withdrawn and awaits repaired evaluation. Simple linear methods can outperform neural methods on bounded endpoints, and a negative result about neural feature-space superiority does not erase the architectural contribution of an explicit, inspectable, and swappable acquisition branch. This manuscript is the first program-level presentation of the complete research program. It is deliberately not a Paired-Acquisition Neural Factorization paper: PA-NF is one of several foundational contributions, and earlier manuscript drafts that treated it as the whole program are rejected here in favor of an accurate, three-level, evidence-maturity-aware structure.

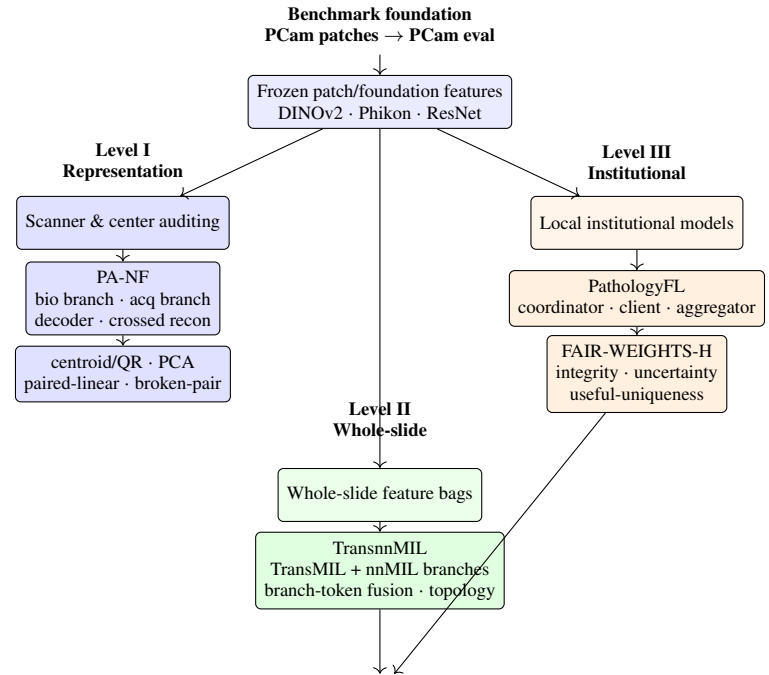
2 Research-program overview

The program is organized as three primary neural aggregation levels plus the benchmark and provenance infrastructure that connects them.

1. **Level I — Representation formation.** How patch and foundation-model representations encode tissue, scanner, stain, acquisition workflow, site or center, region identity, and dataset artifacts. Principal research lines: PCam patch foundation; paired-scanner and paired-acquisition research; Paired-Acquisition Neural Factorization; SCORPION; multi-scanner canine SCC; scanner and center subspaces; centroid/QR, PCA, paired-linear, adversarial, and broken-pair controls; and synthetic identifiability and capacity-allocation studies.
2. **Level II — Whole-slide aggregation.** How patch representations are aggregated into slide predictions. Principal research line: TransnnMIL and all of its implemented architectural families (canonical dual-branch, three-branch v2, branch-token fusion, hierarchical pooling, topology branch, adaptive pruning, graph caching).
3. **Level III — Institutional aggregation.** How

multi-institution updates, validation priorities, contribution signals, and safety constraints are aggregated. Principal research lines: PathologyFL; FAIR-WEIGHTS-H; institutional weighting; dominant-site stress testing; site imbalance; ordinal-grading shift; corruption and label-noise stress; detector behavior; client heterogeneity, dropout, and stragglers; communication accounting; secure aggregation; differential-privacy support; Byzantine or anomaly monitoring; and CAMELYON17 held-out-center and center-weighting studies.

The three levels are connected by a shared evaluation and accountability framework, and by shared frozen feature inputs (DINOv2, Phikon [6], ImageNet ResNet): scanner leakage, center leakage, descriptive task accessibility, region retrieval, whole-slide aggregation, institutional influence, fairness constraints, robustness, provenance, and claim validity are audited with the same fail-closed discipline at every level. PCam provides the patch-level benchmark foundation; PANDA provides the whole-slide benchmark; CAMELYON17 provides the multi-center benchmark; and the provenance system binds every active claim to an immutable artifact. The unifying thesis is that computational-pathology neural systems aggregate information at several levels — patches into representations, patch instances into whole-slide predictions, and institutional updates into collaborative models — and at each level nuisance variation, shortcut signals, optimization failures, and institutional imbalance can be preserved or amplified. This program develops the neural architectures, paired-data designs, institutional-weighting protocols, stress tests, and provenance controls for auditing and improving those aggregation stages.



Cross-cutting: PCam/PANDA/CAMELYON17 benchmarks + provenance + claim validation

Figure 1: Three-level research-program architecture: representation formation (Level I), whole-slide aggregation (Level II), and institutional aggregation (Level III), connected by benchmark and provenance infrastructure. All component boxes are labeled by their evidence maturity in Tables 2–4.

Figure 2: Publication-portfolio map. The flagship foundations manuscript frames the program; focused papers A–D carry scoped evidence. Release order is flagged by evidence maturity.

Component	Scope
Flagship foundations manuscript	all three levels + audit
Focused Paper A (PA-NF)	representation; negative in
Focused Paper B (TransnnMIL)	whole-slide architecture
Focused Paper C (FL/FAIR-WEIGHTS-H)	institutional; infrastructure
Focused Paper D (provenance)	audit system

Evidence-maturity discipline. Throughout, the manuscript keeps four distinct categories separate: *implemented architecture* (binds to source and tests), *corrected empirical evidence* (binds to immutable result artifacts), *negative or mixed results* (reported directly), and *pending controlled validation* (labeled pending). No component is implied to have the same evidence maturity as any other.

3 Benchmark and engineering foundation

Figure 3: Dataset and evidence-boundary map (verified at inventory time).

Dataset	Scale
PCam	96×96 patches; 32,768 test
SCORPION	48 slides, 480 regions, 5 scanners, 2,400 patches
Canine SCC	4,025 patches, 7 categories, 5 scanners
PANDA	10,611 readable Phikon slides, ISUP 0–5
CAMELYON17	455,954 patches, 5 centers

The program is built on patch-level benchmark and evaluation infrastructure that establishes deterministic data handling, evaluation methodology, calibration and uncertainty analysis, and federated smoke validation.

3.1 PCam patch benchmark

PCam (PatchCamelyon) [8, 14] is used as a patch-level engineering and evaluation benchmark: a binary tumor-detection task over 96×96 RGB patches. The implementation includes a dataset API, binary classification head, ResNet and transformer feature extractors, full-config training, full-test evaluation with ROC AUC / accuracy / F1 and bootstrap confidence intervals, stratified cross-validation, threshold operating-point analysis, and failure-asymmetry analysis. Federated smoke and heterogeneity benchmarks use real PCam patches split into simulated sites to exercise weighting strategies.

3.2 What PCam establishes

PCam is treated here as *implementation and benchmark context*, not as an active numerical empirical claim. A single documented single-split run was reported historically, but its underlying numerical result artifacts are not tracked in the repository and are not recoverable, so no specific performance number is promoted as an active cryptographically bound claim in this manuscript or its evidence manifest. What PCam establishes is:

- a working patch-level training and evaluation pipeline (dataset API, heads, feature extractors, training, evaluation with bootstrap CIs);
- deterministic data handling and bootstrap evaluation

- infrastructure with tests;
 - a foundation for later federated and whole-slide infrastructure;
 - a benchmark substrate whose infrastructure, not unbound numbers, is the contribution.
- 3.3 What PCam does not establish
- PCam does not establish pending clinical validation, diagnostic performance, patient-level or slide-level endpoints, external generalization, statistical superiority over unrelated published models, or calibration at a clinically meaningful operating point. The 5-fold cross-validation tables in documentation are projected (the cross-validation run is paused), not observed. The threshold-optimization study is retrospective and was withdrawn as confirmatory or deployment evidence. No selected threshold is recommended for inference or deployment. The current status is *patch-level engineering and evaluation benchmark only*, and clinical and deployment prohibitions are enforced by the authoritative claim boundary.

4 Level I: Paired-acquisition representation formation

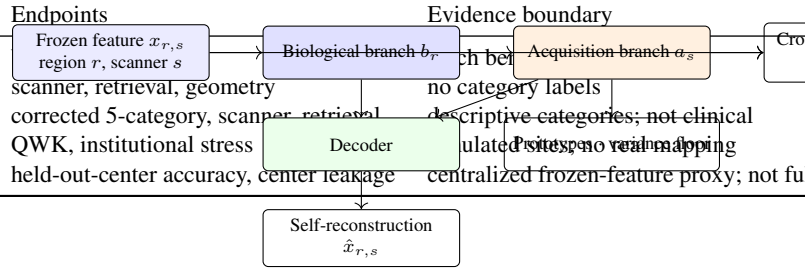


Figure 4: Paired-Acquisition Neural Factorization: an explicit biological branch and acquisition branch are trained on exact same-region multi-scanner pairs with decoder self- and crossed-reconstruction, same-region biological consistency, biological variance floor, and prototype regularization. Category labels never enter factorizer optimization. Corrected evidence supports partial structured separation and paired supervision but not neural feature-space superiority (see Section 4).

Figure 5: Corrected PA-NF findings from the fixed-estimand adjudication v2 (consuming the exactly recovered B32/B64 artifacts). B64–B32 contrasts are means over five folds; the category contrast was positive in one of five folds.

B64–B32 contrast	mean	positive folds	decision
Corrected category BA	–0.008	1 of 5	no category gain
Scanner BA	+0.049	5 of 5	recoverability increases
Worst-pair retrieval	+0.050	5 of 5	retrieval improves
Overall retrieval	+0.051	5 of 5	retrieval improves

Corrected category balanced accuracy (fold 0, seed-averaged): original frozen 0.387; centroid/QR 0.386; paired linear 0.390; B32 0.345; B64 0.326. No neural feature-space increment over every simple scanner-removal baseline was supported; simple baselines are strong.

The first aggregation level is representation formation: how patch and foundation-model representations encode tissue, scanner, stain, acquisition workflow, site or center, region identity, and dataset artifacts. The program’s central design at this level is Paired-Acquisition Neural Factorization (PA-NF), which uses *exact same-region paired multi-scanner scans* to

Figure 6: B32/B64 capacity trade-off under the corrected fixed five-category estimand (seed-averaged means over the recovered cells). Wider biological bottleneck (B64) increases scanner recoverability and region retrieval but does not increase corrected category balanced accuracy.

Endpoint (mean over folds)	B32	B64
Corrected category BA	0.433	0.425
Scanner BA	0.300	0.349
Worst-pair retrieval	0.585	0.634
Overall retrieval	0.719	0.771

Note: values are seed-averaged means over the five folds computed from the adjudication v2 result (canine metric table). The width effect transports to retrieval and scanner recoverability but not to corrected category accessibility.

train an explicit tissue-oriented biological branch and an explicit acquisition branch, together with a decoder and paired cross-reconstruction and consistency objectives. The program also develops the strong simple baselines, broken-pair controls, synthetic identifiability studies, and corrected fixed-estimand evaluation that determine what PA-NF does and does not establish.

4.1 Paired-acquisition design

PA-NF is trained on exact same-region pairs: the same biological region imaged under multiple scanners. Training pairs a source region under one scanner with the same region under another scanner. The biological branch is encouraged to be consistent across scanner views of the same region; the acquisition branch is driven by scanner identity. A decoder reconstructs the target view from the source biological representation and the target acquisition representation (crossed reconstruction). Biological variance floors and prototype regularization shape the branch geometry. The factorizer reads paired region and scanner metadata only; category or biological labels never enter factorizer optimization.

4.2 Datasets

- **SCORPION:** 48 original H&E slides, 480 aligned regions, five scanners, and 2,400 image patches, under original-slide-blocked folds. SCORPION has no validated biological category labels, so only scanner, retrieval, and geometric endpoints are available.
- **Multi-scanner canine SCC:** a multi-scanner external dataset with seven descriptive tissue categories, evaluated under the corrected fixed five-category estimand (Dermis, Epidermis, Inflamm/Necrosis, SCC, Subcutis; Bone and Cartilage excluded) with biological-sample-blocked folds, fit-only probe standardization, and same-region and same-sample nearest-neighbour exclusion.

4.3 Corrected empirical evidence

The corrected July 26 forward-valid evidence release supports *partial structured separation under the tested conditions*, not pure biological factors, complete independence, or clinical utility. Corrected paired-scanner experiments demonstrate paired supervision relative to broken-pair controls and support retrieval and scanner-recoverability findings.

The versioned fixed-estimand adjudication (v2, consuming the

exact recovered neural artifacts) established the current status `complete_no_neural_feature_space_increment_supported`. Under the corrected five-category estimand, on the recovered B32/B64 cells:

- B64 *did not* improve corrected five-category canine category balanced accuracy (B64 minus B32 mean -0.008 , one positive fold).
- B64 *did* increase scanner recoverability (mean $+0.049$, five of five folds) and improve worst-pair and overall region retrieval (mean $+0.050$ and $+0.051$, five of five folds).
- No neural feature-space increment over every simple scanner-removal baseline was supported: neither B32 nor B64 achieved a category gain of 0.02 over all simple baselines in four of five folds, and both violated the worst-pair retrieval margin.
- Simple baselines (centroid/QR, paired linear) are strong: they reach category and retrieval levels the neural branches do not, while the neural biological branch can reach lower scanner recoverability in some folds. Neither a neural increment nor a strict simple-baseline Pareto dominance was established.
- Synthetic accessibility effects did not transport: the corrected category gain required for transport was not present.

4.4 Simple baselines and controls

The program treats simple baselines seriously. Centroid/QR scanner-subspace projection, PCA scanner-component removal, and paired linear scanner transforms are all recomputed from the same frozen arrays under the same candidate pool and probe family. Oldstyle centroid/QR is the strongest raw scanner-removal baseline (original frozen-feature scanner probe ≈ 0.85 reduced to ≈ 0.20 at chance in corrected audits). Broken-pair controls preserve scanner counts and remain within the training fold while deranging region correspondences, isolating the contribution of true pairs.

4.5 Why the architecture remains distinct despite the absence of superiority

The negative neural-increment result does not erase the architectural contribution. PA-NF produces an explicit, inspectable, and decoder-swappable acquisition branch and a biological branch whose scanner recoverability can be reduced and whose structure can be bottlenecked. Centroid/QR is strong at blind removal but provides no structural decomposition. The architecture is a distinct structured method whose empirical value on corrected bounded endpoints is a negative or mixed result — and that result is reported directly in the main text rather than hidden.

4.6 Layer-2 boundary

Validated scanner swapping through the decoder (Layer 2) is *not* established: verified source-region, target-scanner, and swap-assignment metadata are absent. Swap assignments are never inferred from filenames, row order, or scanner labels. The missing-metadata schema is a future data-remediation specification only. Pixel-space evaluation and clinical interpretation are prohibited by evidence scope.

5 Level II: TransnnMIL whole-slide aggregation

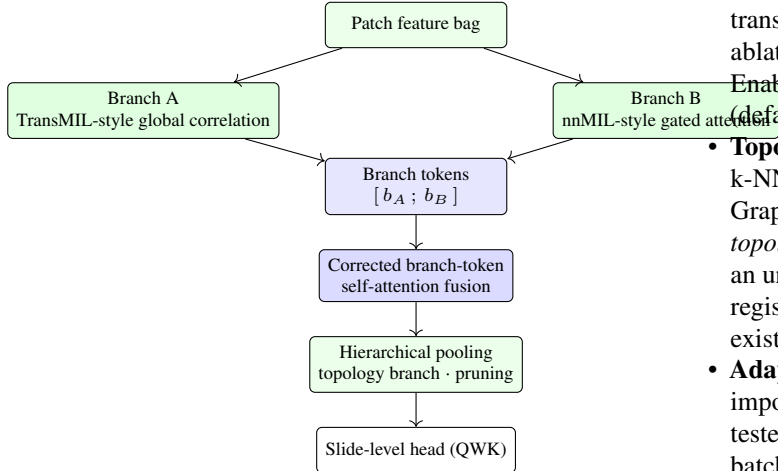


Figure 7: TransnnMIL architecture family. The repaired canonical model fuses explicit branch tokens by self-attention. Historical fusion and topology results are withdrawn; post-repair PANDA runs are preliminary and pending matched controlled reruns (see Section 5).

Whole-slide prediction requires aggregating tens of thousands of patch representations into a single slide-level decision. This is the second primary aggregation level. The program’s contribution here is TransnnMIL, a multibranch whole-slide neural aggregation architecture family. The architecture is an authored and implemented contribution; its corrected matched-control superiority evaluation is pending. Historical performance numbers produced by a defective execution path are withdrawn and are never used as repaired-model evidence.

5.1 Verified architectural families

The implementation lives in `src/models/transnnmil/` and is described here from the code, not from stale marketing prose. The verified families are:

- **Canonical TransnnMIL** (`transnnmil.py`): a dual-branch model with a TransMIL-style global-correlation branch (`src/models/mil/transmil.py`, after TransMIL [12]) and an nnMIL-style gated-attention branch (`src/models/mil/nnmil.py`, after attention-based multiple-instance learning [7]). The repaired model fuses the branches by *self-attention over explicit branch tokens*: the two projected branch representations are stacked and attended (`branch_token_fusion.py`), replacing the withdrawn one-query one-key cross-attention path that was query-invariant. Explicit branch representations are implemented and tested.
- **TransnnMILv2** (`transnnmil.v2.py`): a three-branch variant (TransMIL + hierarchical pooling + topology), a two-branch variant with AB/AC/BC ablations, and explicit branch-token fusion. No empirical results exist for these variants; numbers in design documentation are explicitly projected, not observed.
- **Fusion controls** (`branch_token_fusion.py`, `transnnmil_branch_token.py`): learned branch-attention, concat, and gate fusion operators, aligned to the repaired canonical branches. Implemented and tested, but *planned controls* with no recorded results yet.

- **Hierarchical spatial pooling** (`hierarchical_pooling.py`): learnable cluster centers, region attention/mean/max pooling, and region transformers. Implemented and tested; a synthetic-MIL ablation exists, and a real-data benchmark is planned. Enabled in the canonical model only when requested (default off); PANDA runs did not enable it.
- **Topology or graph branch** (`topology_branch.py`): k-NN graph construction and GNN layers (GATv2, GraphSAGE, GIN). Implemented and tested. *Historical topology results are withdrawn*: the old forward pass created an unregistered random projection. The repaired model registers the projection; a post-repair rerun does not yet exist.
- **Adaptive pruning** (`adaptive_pruning.py`): importance scoring and adaptive pruning. Implemented and tested, but disabled in TransnnMILv2 pending a proper batched implementation; no results.
- **Graph caching and coordinate-aware processing** (`graph_cache.py`): HDF5 k-NN graph precompute and cache. Implemented and tested; no training result.

5.2 Branch-token fusion mechanism

The corrected fusion is the technical heart of the canonical model. Each branch produces a sequence of patch-level representations; the branch representations are projected to a fixed token dimension and stacked into an explicit branch-token sequence $T \in \mathbb{R}^{2 \times d}$, one token per branch. A multi-head self-attention over T computes the fused slide representation as a learned interaction between the two branch tokens, and the attended tokens are mean-pooled before the classifier head. This fixes the historical defect: the old one-query/one-key cross-attention used a single query token against a single key, so the softmax over one position was identically 1 and the fusion was *query-invariant* — it could not weight the branches at all. The repaired design gives the fusion genuine branch-interaction capacity, and regression tests require output sensitivity and gradient flow to both branches (`tests/models/test_transnnmil_canonical_fusion.py`). The incremental contribution of this fusion relative to concat and gate controls is *not yet measured*; that is exactly the pending matched-control evaluation. A full per-component implementation/test/result matrix is in the supplement.

5.3 PANDA training infrastructure and feature bags

The PANDA training infrastructure uses a verified Phikon 768-dimensional feature bag manifest (10,611 readable slide feature files; 8,488 train and 2,123 validation rows) with `train_panda_transnnmil_baseline.py`. Stability grids across learning rates and seeds show best validation QWK in the 0.812–0.826 range, and ablation runs exist. These numbers are *preliminary and pending*: they are not yet matched controlled comparisons against TransMIL, nnMIL, AttentionMIL, concat, gate, and learned branch-attention controls under identical splits, seeds, tuning budgets, and compute.

5.4 Evidence status

- **Implemented and tested architecture:** all families above have unit and integration tests.
- **Active empirical evidence:** PANDA stability and ablation runs on the repaired canonical model are *preliminary*, not superiority evidence.
- **Withdrawn evidence:** historical fusion and topology results, including historical QWK values, are withdrawn by the authoritative claim boundary. They remain records of the old defective execution path.
- **Pending:** repaired matched controlled reruns (the ablation plan in `docs/results/panda-transnnmil-ablation-plan.md`) are required before any superiority claim.

The architecture remains a distinct authored contribution and an active research line even though its corrected superiority evaluation is pending. Projected AUC values in design documents are never treated as observed results, and historical QWK values are never reused as repaired-model evidence.

6 Level III: PathologyFL

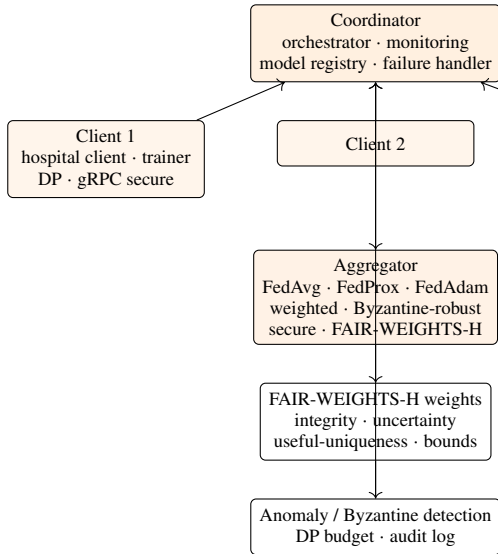


Figure 8: PathologyFL coordinator–client architecture with the FAIR-WEIGHTS-H weighting layer and anomaly/monitoring paths. Implemented research infrastructure; no real multi-center clinical deployment exists (see Section 6).

The third aggregation level is institutional aggregation: how multi-institution updates, validation priorities, contribution signals, and safety constraints are aggregated into a collaborative model. The program’s contribution here is PathologyFL, a pathology-specific federated-learning research framework, and FAIR-WEIGHTS-H, an auditable hybrid institutional-weighting protocol.

6.1 Architecture

PathologyFL implements a coordinator-client architecture under `src/features/federated/pathology_fl/`:

- **Coordinator:** a production orchestrator with round lifecycle, client selection, Byzantine filtering, a tamper-evident audit-log hash chain, and versioned checkpoints; a failure handler with timeouts, retries, blacklisting, and recovery; monitoring (Prometheus,

TensorBoard, convergence detection, alerts); and a model registry with SHA-256 integrity and rollback.

- **Client:** a production hospital client with local training, differential-privacy integration [1], gRPC secure communication, and round participation; a local trainer; a resource manager; and a PACS connector.
- **Aggregator:** FedAvg [11], FedProx [10], FedAdam, an explicit weighted adapter (for external FAIR-WEIGHTS-H weights), Byzantine-robust aggregation (Krum [3], multi-Krum, trimmed mean, median) with a Byzantine detector, a pathology-aware adapter, a secure aggregator [4], and a FAIR-WEIGHTS-H aggregator.
- **Privacy:** differential-privacy engines (including an Opacus-backed DP-SGD engine), privacy budget tracking and accounting, and noise calibration.
- **Monitoring and anomaly detection:** coordinator monitoring plus Byzantine detection and FAIR-WEIGHTS-H per-parameter-group risk and reason codes.
- **Secure communication and aggregation:** TLS-based gRPC with certificate validation and mutual auth, and a TenSEAL-backed secure aggregation protocol that rejects plaintext in the aggregate-only path.
- **Heterogeneity and systems concerns:** FedProx for non-IID proximal terms, async training with staleness weighting and adaptive timeouts, fault tolerance with checkpoint management, network monitoring, and reconnection handling, compression (quantization, sparsification), and client resource awareness.

6.2 Round lifecycle and failure handling

The orchestrator runs a deterministic round lifecycle: select clients, distribute the global model, wait for updates (with adaptive per-client timeouts), apply Byzantine filtering and aggregation, and write a tamper-evident audit-log entry whose hash chain covers the round’s client set, received updates, and aggregate. A client that times out or returns a non-finite update is retried, blacklisted after repeated failure, and recovered through a checkpointed round resume. Async modes weight stale updates by an exponential staleness decay. This lifecycle is what makes dropout and straggler behavior an *accountable* part of the framework rather than an implicit assumption: it is tested by the dropout and checkpoint-recovery integration tests.

6.3 Execution validation

PathologyFL has integration and property tests that exercise five-client training, convergence, DP budget enforcement, Byzantine attacks, client dropout and checkpoint recovery, weighted aggregation, secure-aggregation homomorphism, and compression. A PCam-derived federated smoke run uses real PCam patches split into five simulated sites and four weighting strategies.

6.4 Evidence status

The framework is *implemented research infrastructure* with validated smoke and integration behavior. It is *not* a validated multi-center deployment system. Key distinctions are enforced:

- implemented infrastructure and tested execution are separate from simulated pathology benchmarks;
- no real multi-center clinical deployment exists;
- PCam smoke results are plumbing validation, not performance evidence;
- centralized frozen-feature CAMELYON17 studies are not described as full federated-learning validation.

Caveats are documented. The FAIR-WEIGHTS-H aggregator integration (`pathoalign_fair.py`) has no dedicated test file. Differential-privacy (Opacus-backed DP-SGD) and secure-aggregation (TenSEAL-backed) paths require optional libraries; in an environment without those libraries the corresponding tests skip cleanly and the guarantees are not independently audited. The integration, aggregation, production-integration, privacy-regression, and FAIR-WEIGHTS-H engine tests pass; the *e2e* federated tests are gated as slow and collect cleanly.

7 FAIR-WEIGHTS-H

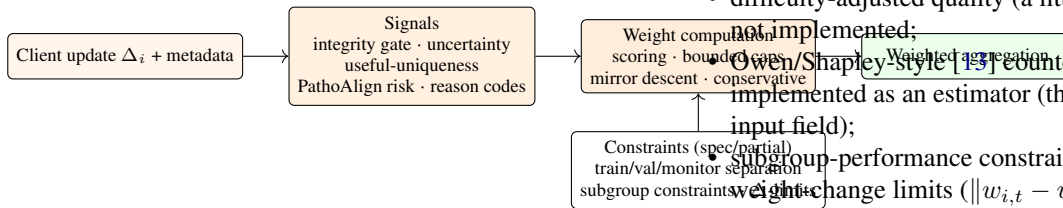


Figure 9: FAIR-WEIGHTS-H signal and constraint flow. Implemented: integrity gate, uncertainty penalty, useful-uniqueness, bounded weights, PathoAlign audit bridge, conservative mode. Specification-only or partial: training/validation/ monitoring weight separation, difficulty adjustment, Shapley-style contribution, subgroup constraints, weight-change limits. No established performance or fairness superiority (see Section 7).

FAIR-WEIGHTS-H is an auditable hybrid institutional-weighting protocol proposed and partially implemented under PathologyFL. It is designed to aggregate institutional updates using a bounded, auditable weight per institution rather than sample volume alone. The protocol is implemented or specified as follows (verified from the implementation and its specification):

7.1 Implemented and tested

- **Weight engine** (`weighting/fair_weights.h.py`): linear and log-linear scoring, softmax and mirror-descent updates, caps and renormalization, normalized entropy and effective-institution-count diagnostics, an integrity gate (a failing integrity check drives a weight toward zero), an uncertainty penalty, and a conservative mode. These are implemented and unit-tested.
- **PathoAlign audit bridge** (`weighting/pathoalign_bridge.py`): derives an adjusted-quality score, a useful-uniqueness signal, and an uncertainty penalty from validation metrics and nuisance/center audit signals, with reason codes and a mechanism-validity flag.

7.2 Weight-computation mechanism

The implemented engine scores each institution from a small signal set: an integrity gate (a failing integrity check drives the score to -10^9 , so the softmax weight is effectively zero), an

uncertainty penalty, and a useful-uniqueness term. Scores are converted to weights by a softmax or a mirror-descent update that moves the weight vector toward the target scores subject to caps and renormalization; normalized entropy and an effective-institution-count diagnostic quantify how concentrated the influence is. The PathoAlign audit bridge maps validation AUC / balanced accuracy / calibration plus nuisance and center audit signals into an adjusted quality, a useful-uniqueness term, a representation-risk value, and reason codes with a mechanism-validity flag. A conservative mode and an aggregator fallback-to-equal path bound the effect when signals are unreliable. The full signal and constraint definitions are in the supplement.

7.3 Specified but not fully implemented

Several protocol elements are specified in the design documentation but are not implemented or are only partial:

- separation of training, validation, and monitoring weight vectors ($w_{\text{train}} \neq w_{\text{val}} \neq w_{\text{monitor}}$) is specification-only;
- difficulty-adjusted quality (a fitted case-difficulty model) is not implemented;
- Owen/Shapley-style [13] counterfactual contribution is not implemented as an estimator (the contribution score is a free input field);
- subgroup-performance constraints and explicit weight-change limits ($\|w_{i,t} - w_{i,t-1}\| \leq \Delta$) are specification-only;
- the four-mode Normal/Conservative/Safety/Emergency fallback table is specification-only, though a conservative mode and a fallback-to-equal aggregator path exist;
- validated automated drift detection and fallback selection are not implemented.

7.4 Execution validation

Execution stability and aggregation behavior have been tested: synthetic perturbation experiments, a PCam federated smoke (all four weighting strategies achieve similar ≈ 0.53 accuracy, confirming plumbing rather than performance), and PANDA-derived simulated-federation stress tests. In the PANDA-3000 simulated stress, FedAvg is strongest in the clean regime, a cross-site blend is best at 25% dominant-site label noise, and results are mixed at higher noise. These are *conditional, simulated-site* results.

7.5 Exact claim status

Consistent performance or fairness superiority over simpler baselines is *not* established by any current immutable artifact. FAIR-WEIGHTS-H results live under `results/` without forward-valid release manifests. The documented boundary is preserved exactly:

- execution stability and aggregation behavior may be supported;
- consistent performance or fairness superiority over simpler baselines is not established;
- “FAIR” in the method name does not mean fairness has been proven;
- passing software tests does not establish improvement on real multi-institution data, clinical validity, fairness across patient groups, privacy guarantees, or deployment readiness;

- prospective and multi-center validation remain necessary.

8 PANDA institutional and ordinal studies

PANDA (Prostate cANcer graDe Assessment) [5] provides the whole-slide ordinal grading benchmark (ISUP 0–5, quadratic-weighted kappa as the primary metric) and the substrate for institutional-aggregation stress experiments. The program’s PANDA studies are organized into feature-bag and slide aggregation, ordinal grading, institutional weighting, dominant-site stress, and corruption and label-shift experiments.

8.1 Feature bags and slide aggregation

A verified Phikon feature-bag manifest covers 10,611 readable slide feature files (3 unreadable files dropped; 8,488 train and 2,123 validation rows). Slide-level baselines are mean-pooled Phikon + MLP (QWK 0.7274) and gated AttentionMIL (QWK 0.8100). The repaired TransnnMIL canonical model reaches best-validation QWK 0.8155 (tuned seed), with repeated seeds 0.8225 and 0.8086, and an 18-run stability grid over learning rates and seeds spanning ≈ 0.812 – 0.826 . *These are preliminary, not matched-controlled superiority results;* historical fusion and topology values are withdrawn.

8.2 Institutional weighting

FAIR-WEIGHTS-H, cross-site contribution blends, and FedAvg are compared on PANDA-derived simulated federations. FedAvg is strongest in the clean regime; contribution-aware blends are best at 25% dominant-site label noise; results are mixed at higher noise. These are simulated-site, cached-feature experiments over synthetic institutional partitions, not real hospital federations.

8.3 Dominant-site stress, ordinal shift, and detector behavior

Simulated dominant-site label-flip corruption (0–45%) over five simulated sites tests when sample-volume dominance becomes unsafe under controlled corruption. Ordinal-harm-aware analysis examines grading-shift failures. A tuned dominance detector reduces clean false-switch from 33.3% to 6.7% and shows significant global-QWK gains at 25%/35% noise over 15 seeds; detector transfer from label-noise calibration to ordinal threshold shift and detector ablation are documented.

8.4 Centralized versus federated

A centralized-vs-federated comparison on the 1,000- and 3,000-slide Phikon caches shows centralized $>$ FedAvg $>$ local-only ordering (e.g., global QWK 0.6949 vs 0.6659 vs 0.6075 at 3,000 slides). This is simulated-site, cached-feature evidence only.

8.5 Patient-mapping and evaluation limitations

PANDA provides slide-level ISUP grades. The caches are mean-pooled slide vectors; sites are synthetic splits over cached feature subsets and do not correspond to a real hospital federation. No patient-level endpoint, real hospital mapping, or clinical validity is claimed.

9 CAMELYON17 center studies

CAMELYON17 [2] provides a multi-center whole-slide benchmark, structured here with the WILDS protocol [9] (five centers, 455,954 labeled patches) and is used to study source-center weighting, center leakage, and held-out-center generalization. The program’s CAMELYON17 work is organized as centralized feature-level proxies and representation-mechanism diagnostics, and is explicitly *not* full federated-learning validation.

9.1 Source-center weighting

Three aggregation-weighting policies are compared on held-out-center accuracy: sample-proportional (FedAvg-style) patch weighting, equal-client weighting, and downweighting the dominant center. On frozen ImageNet ResNet18 features, equal-client weighting reaches 0.9132 held-out accuracy versus 0.8312 for sample-proportional (+8.20 pp); on supervised ResNet18 features, 0.9318 versus 0.9052 (+2.66 pp). The WILDS structure uses source centers $\{0,3,4\}$ for training and in-distribution validation, center 1 as the out-of-distribution validation center, and center 2 as the held-out test center. Detector-switch decisions use only in-distribution validation diagnostics, never the held-out test center.

9.2 Center subspace and center leakage

Adversarial and subtractive nuisance-removal variants (v3–v5) did not materially reduce post-hoc center leakage (a negative result). Supervised linear center-subspace projection (v6c/v7) reduces center decodability (center accuracy $0.895 \rightarrow 0.764$ for five centers at effective rank four) while tumor AUC remains ≈ 0.990 . These results are active only as representation-mechanism diagnostics.

9.3 Pillar studies

The four-pillars framework organizes federated-learning failure modes: non-IID site shift (strongest evidence), communication overhead (a bounded feature/head resolution for the communication-accounting pillar), residual privacy risk (a coefficient-noise robustness probe), and implementation and infrastructure barriers (an infrastructure-friction simulation with client speed, dropout, and straggler accounting).

9.4 Centralized proxies versus full federated learning

The CAMELYON17 weighted logistic-regression studies are centralized frozen-feature proxies for source-center influence on one held-out center. They do not validate a complete federated-learning algorithm, and one held-out center does not establish a general law across hospitals. The planned Track A validation (centralized baseline, FL baselines, detector transfer) is planned, not completed. FAIR-WEIGHTS-H has no real CAMELYON17 validation; the synthetic CAMELYON17 smoke is explicitly historical and superseded.

10 Synthetic and mechanism studies

Synthetic and controlled mechanism studies are organized by the question they test rather than by dataset. Because the latent biological and acquisition factors are known in synthetic generators, recovery can be measured rather than inferred

indirectly. All synthetic claims are kept strictly separate from pathology claims.

10.1 Identifiability

Synthetic crossed-factor identifiability studies ask when ordinary observations and paired interventions jointly control recovery of latent biological and acquisition factors, covering observational sample size, unique paired-anchor count, factor overlap, linear versus nonlinear mixing, and seed. Crossed-target scanner-prototype factorization and unseen-identity representation-geometry studies examine factor recovery under crossed targets and held-out identities. Finite-sample whitening identifiability audits examine when whitening aids or confounds recovery.

10.2 Factor allocation and capacity

Paired-acquisition dimensionality $\times$ cross-covariance factorials (450 cells) and capacity-matched campaigns (175 fits) probe the separation frontier of biological and acquisition branches. The biological-bottleneck capacity allocation factorial (status `complete_capacity_gain_with_scanner_tradeoff`) showed a capacity gain with a scanner tradeoff. Routed paired-consensus bottleneck and paired-consensus linear-anchor studies test routed and consensus supervision. The real paired-scanner bottleneck-allocation validation (`complete_mixed_real_paired_scanner_allocation`) transferred the capacity question to real paired-scanner data: B64 improved retrieval and increased scanner recoverability without a category gain.

10.3 Task sufficiency

Task-defined biological-sufficiency benchmarks and task-benchmark instrument-power audits measure how much capacity is actually used by a task and whether the evaluation instrument has power to detect differences. Minimal and whitened biological-bottleneck studies and residual-probe nonlinear-capacity calibration examine how representation geometry responds to constrained capacity.

10.4 Synthetic-to-real transport

Synthetic accessibility effects do not automatically transport to real pathology features. The corrected fixed-estimand adjudication established that the synthetic width effect did not produce a corrected real category gain:

`synthetic_accessibility_effect_transport` is false because B64 did not improve corrected five-category canine balanced accuracy. Retrieval gain is never counted as transport of biological accessibility.

10.5 Interpretation boundary

Synthetic results demonstrate mechanism, identifiability, and allocation behavior under controlled generators. They are not pathology evidence, do not establish clinical utility, and are never used to inflate a real-data claim beyond what the corrected real-data artifacts support.

11 Cross-level accountability framework

The three aggregation levels are connected by a shared accountability framework: the same fail-closed discipline is

applied to auditing scanner leakage, center leakage, descriptive task accessibility, region retrieval, whole-slide aggregation, institutional influence, fairness constraints, robustness, provenance, and claim validity. This section is the conceptual glue of the manuscript.

11.1 Audit endpoints at each level

- **Representation level:** scanner and acquisition recoverability (linear and nonlinear probes with paired permutation nulls), descriptive tissue-category accessibility under the corrected fixed-estimand, region retrieval and same-region cosine similarity, acquisition-category leakage, and spectral accessibility.
- **Whole-slide level:** slide-level ordinal metrics (QWK), matched controlled comparisons across MIL architectures, computational efficiency and stability, and branch-contribution attribution.
- **Institutional level:** institutional influence weights, dominant-site and corruption stress, ordinal-shift detection, held-out-center accuracy, privacy and secure-aggregation behavior, and communication accounting.

11.2 How artifacts propagate across levels

Scanner and center shortcuts can propagate from representation features into MIL aggregation and federated updates. The framework makes the propagation checkable: a frozen feature array can be audited for scanner and center recoverability before it enters a whole-slide model or a federated client. An explicit acquisition or institution branch can be bottlenecked, inspected, and (in the presence of verified swap metadata) swapped, which a blind linear projection cannot provide.

11.3 Claim validity as an audit artifact

Every active numerical claim binds to an immutable result artifact with a recorded hash; every architectural claim binds to source and tests; every theoretical or protocol claim binds to its exact specification file. Negative results are reported in the main text. Pending controlled validation is labeled pending. This framework is what allows the manuscript to present a negative neural-increment result, a pending TransnnMIL superiority evaluation, and a partially specified institutional-weighting protocol without conflating their evidence maturity.

12 Scientific audit and provenance

A central methodological contribution of the program is a fail-closed scientific-audit and provenance system that separates code correction from empirical validation and binds every active claim to an artifact that can be cryptographically re-verified. This is a cross-cutting research-engineering contribution, not administrative filler: it is what makes the negative and pending statuses reported in this manuscript trustworthy.

12.1 Immutable evidence releases

Three separately versioned, immutable evidence packages live under `evidence/paired_acquisition/`:

- `corrected-20260726` — the corrected forward-valid

Figure 10: Current evidence-status matrix (2026-08-04). Architecture, corrected empirical evidence, negative results, and pending validation are kept separate.

Research line	Architecture	Corrected empirical
PA-NF	implemented	partial separation; paired supervision
TransnnMIL	implemented	preliminary PANDA runs
PathologyFL	implemented	integration/smoke validation
FAIR-WEIGHTS-H	partial	execution validation
PCam	implemented	observed test-split run
PANDA studies	implemented	simulated-site results
CAMELYON17	implemented	centralized proxies; mechanism
Synthetic	implemented	mechanism evidence
Provenance/audit	implemented	validated by its own tests

- paired-acquisition release (canine fixed-estimand + SCORPION fold-aware evidence), validated by `validate_corrected_paired_acquisition_evidence.py`.
- dimensionality-xcov-factorial-20260726 — the 450-cell dimensionality $\times$ cross-covariance factorial, validated by `validate_paired_acquisition_factorial_evidence.py`.
- scorpion-capacity-matched-20260726 — the 175-fit capacity-matched campaign, validated by `validate_scorpion_capacity_matched_evidence.py`.

Each release carries a manifest that binds source-code commits (with tree-equivalence checks), promoted artifacts (SHA-256 and byte size), row counts, and a claim-boundary snapshot. Validators recompute declared metrics and fail closed on any mismatch. The corrected release also implements a living claim-boundary contract: the immutable snapshot’s canonical-text hash equals the release-time publication hash (which can never change), while the current authoritative claim-boundary file is a living document whose hash is reported informationally.

12.2 Historical withdrawal and remediation

The authoritative claim boundary records which claims were withdrawn and why. The 2026-07-25 remediation ledger (`docs/research/scientific-audit-remediation-20260725.md`) documents corrected defects — for example, TransnnMIL’s query-invariant attention and unregistered topology projection, canine test-set leakage and scale-incomparable probes, SCORPION slide-level sign-flip withdrawal, and PCam overclaims — and distinguishes contaminated-protocol claims from valid corrected evidence. Withdrawn claims may be discussed as motivation for remediation but never appear in active result tables.

12.3 Deterministic replay and exact artifact recovery

The real paired-scanner run originally persisted metrics but no per-cell representations, leaving the fixed-estimand adjudication at `fixed_estimand_adjudication_not_ready`. An exact deterministic replay (`run_real_bottleneck_representation_recovery.py`) re-executed the 50-cell canine B32/B64 grid on CUDA using the exact frozen training implementation (source-equivalence verified against the frozen commit), compared every cell against its frozen per-run record under a pre-fixed 10^{-6} tolerance, and archived `projected_features.npz` and

`checkpoint.pt` for all 50 accepted cells. This is provenance remediation, not a new experiment.

12.4 Fail-closed statuses

Pending / boundary Every runner uses a strict status vocabulary. Recovery uses `complete_exact_real_bottleneck_representation_recovery` versus `real_bottleneck_representation_recovery_failed`. The fixed-estimand adjudication uses a scientific vocabulary (complete, no neural feature space increment supported etc.) and reserves `no pathology transport` for integrity failures. Poor scientific results receive scientific statuses; execution failures receive failure statuses. Claim verdicts are supported, unsupported, or prohibited by evidence scope.

12.5 Why this materially affects scientific reliability

The audit system makes the distinction between implemented architecture, corrected empirical evidence, negative results, and pending controlled validation enforceable rather than stylistic. A reader can verify that an active numerical claim binds to an immutable artifact with a recorded hash, that a withdrawn result is not reused, and that a pending claim is labeled pending. The negative neural-increment result and the pending TransnnMIL superiority evaluation reported in this manuscript are themselves products of this system: they are the outcome of fail-closed, reproducible evaluation, not of selective reporting.

13 Discussion

13.1 Aggregation choices interact across levels

Representation formation, whole-slide aggregation, and institutional aggregation are not independent engineering problems. A scanner shortcut encoded in patch representations can be amplified by a whole-slide attention model and then propagated into federated updates. Conversely, an institutional-weighting constraint can change which local models contribute and therefore which representation artifacts dominate the aggregate. The program treats these interactions as the object of study.

13.2 Architecture novelty and superiority are different

An architecture can be an authored, implemented, and tested contribution even when its superiority over matched baselines is unproven or has been withdrawn pending repair. TransnnMIL is an implemented multibranch architecture whose corrected matched-control evaluation is pending; PA-NF is an implemented paired factorization whose corrected empirical value is a negative or mixed result on bounded endpoints. Neither conclusion erases the architectural contribution.

13.3 Simple methods can outperform neural methods on bounded endpoints

On corrected bounded endpoints, simple scanner-removal baselines such as centroid/QR and paired linear transforms are strong: they can match or exceed neural category and retrieval levels while the neural biological branch can reach lower

scanner recoverability in some folds. This is a positive methodological result: bounded evaluation instruments must include strong simple baselines, and a neural method’s value may lie in structure (inspectable, bottleneckable, swappable branches) rather than in raw endpoint superiority.

13.4 Negative results belong in the main text

The corrected fixed-estimand adjudication found no neural feature-space increment; synthetic accessibility did not transport; adversarial center-removal did not reduce center leakage. These negative results are reported directly because they bound the program’s claims and motivate the required next experiments (matched TransnnMIL reruns, prospective FAIR-WEIGHTS-H validation).

13.5 Explicit branches can remain scientifically useful

An explicit acquisition or institution branch provides structure — an inspectable representation that can be bottlenecked, constrained, and, with verified swap metadata, swapped. Structural difference from a projection baseline is real even when empirical superiority is absent, but structural difference is not by itself empirical superiority.

13.6 Shortcut propagation into MIL and FL

Scanner and center shortcuts can propagate from frozen features into MIL and federated models. The cross-level accountability framework makes this checkable at every stage, and the provenance system makes the checks reproducible.

13.7 What requires external validation

Repaired matched TransnnMIL reruns, prospective and multi-center FAIR-WEIGHTS-H validation, real multi-center PathologyFL deployment, and validated Layer-2 scanner swapping all require external evidence beyond the current corrected artifacts.

13.8 Which components are ready for focused papers

The publication portfolio identifies the flagship foundations manuscript and focused papers for PA-NF, TransnnMIL, PathologyFL/FAIR-WEIGHTS-H, and the scientific-audit system, with each paper’s evidence maturity stated explicitly.

14 Limitations

The program’s claims are bounded by the following limitations, all of which are reflected in the evidence manifest and claim ledger:

- **TransnnMIL:** repaired matched controlled reruns are incomplete; historical fusion and topology results are withdrawn; projection numbers in design documents are not observed results; adaptive pruning is disabled and hierarchical pooling lacks a real-data result; several v2 variants and fusion controls have no empirical results.
- **Paired acquisition:** there is one labeled paired multi-scanner category dataset (canine SCC); SCORPION has no validated category labels; evidence is frozen-feature only; validated Layer-2 scanner swapping is absent (no verified swap metadata); no pixel-space validation; no clinical deployment; the corrected fixed-estimand adjudication supports no neural feature-space increment and no synthetic accessibility transport; simple baselines are

strong; fold counts are limited.

- **PathologyFL:** no real multi-center deployment; e2e federated tests have dangling imports; secure-aggregation and DP guarantees are gated on optional libraries and not independently audited; the FAIR-WEIGHTS-H aggregator integration is untested.
- **FAIR-WEIGHTS-H:** no proven advantage over simpler baselines; several protocol elements (training/validation/monitoring weight separation, difficulty adjustment, Shapley-style contribution, subgroup constraints, weight-change limits, full fallback modes) are specification-only or partial; “FAIR” does not imply proven fairness.
- **PANDA and CAMELYON17:** simulated institutional sites over cached feature subsets; no patient-level endpoint or real hospital mapping; limited held-out centers (one CAMELYON17 held-out center); centralized CAMELYON17 studies are frozen-feature proxies, not full federated learning.
- **PCam:** patch-level benchmark only; not clinical validation; cross-validation paused; threshold study withdrawn.
- **General:** no absolute priority claim is made without a current external literature review; synthetic-to-real transport failures are documented; all pathology claims are bounded to descriptive endpoints and are never clinical endpoints.

These limitations are stated in the main text rather than deferred to a supplement because they materially shape which conclusions may be drawn at each level.

15 Conclusion

This research program develops accountable neural aggregation in computational pathology across three connected levels: representations (how patches and foundation models encode tissue, scanner, acquisition, site, region, and dataset artifacts), whole slides (how patch instances are aggregated into slide predictions), and institutions (how institutional updates are aggregated into collaborative models). The program’s contributions span neural architectures (Paired-Acquisition Neural Factorization and TransnnMIL), paired-data designs (exact same-region scanner pairs and their controls), institutional-weighting protocols (PathologyFL and FAIR-WEIGHTS-H), stress tests and benchmarks (PCam, PANDA, CAMELYON17), and a fail-closed provenance and scientific-audit system that binds every active claim to an immutable artifact.

The evidence maturity is heterogeneous and is stated explicitly. Corrected paired-scanner evidence supports partial structured separation and paired supervision but not neural feature-space superiority; the corrected fixed-estimand adjudication found no neural feature-space increment, and synthetic accessibility effects did not transport. TransnnMIL is an implemented architecture whose corrected matched-control performance evaluation remains pending. PathologyFL and FAIR-WEIGHTS-H provide executable, auditable federated research and institutional weighting without demonstrated clinical or fairness superiority.

The shared contribution is a multi-level framework for building and auditing accountable neural aggregation in

computational pathology — one in which architecture, corrected empirical evidence, negative results, proposed protocols, and pending controlled validation are kept rigorously distinct, and in which the aggregation choices at each level are treated as a connected system rather than separate engineering problems.

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